

YAJAT KUMAR

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Experience

Routing/IP Software Engineer 2 – Ciena Ottawa, ON

Feb 2024 - Present

- Engineered autonomous AI agents enabling GDB debugging and packet capture analysis inside live Docker containers, reducing mean debug time by 57%.
- Developed fault-tolerant distributed control-plane software, scaling modular Docker-based microservices across 41 platforms spanning ARM and x86 architectures.
- Contributed to a high-scale, multi-threaded C redesign of path computation engine spanning 10,000+ nodes and 410K+ routes, improving convergence time by 31%.
- Python development of high-throughput network validation tests, leveraging parallelization and batching to reduce integration testing cycle times by 43%.

Routing/IP Software Engineer 1 – Ciena Ottawa, ON

Feb 2023 - Feb 2024

- Developed and maintained critical control-plane routing software for high-scale optical networking platforms supporting up to 32 Tbps of network capacity.
- C/C++ development of core software features, resolving 13 critical system-level concurrency bugs and reducing code regression defects by 17% within performance-critical components.
- Contributed to Segment Routing (SR) control-plane infrastructure, engineering high-availability mechanisms to guarantee robust state preservation and seamless system failover in under 50ms.
- Implemented centralized path-computation and policy association features (PCEP / SR Policy) complying with RFC 9862 and RFC 8281 to enable dynamic traffic steering across topologies scaling up to 8,000+ distributed network nodes in Software-Defined Networking (SDN) architectures.

Projects

Pear Launcher | Java, MySQL

Nov 2017 - Present

- Android home screen replacement app with 1M+ installs and 25K+ active users.

Android OS Port for Arduino Uno Q and OrangePi Zero 2W & 3 SBC | Embedded Linux / AOSP

- Ported Android custom ROM on upstream Linux 6.18 Generic Kernel Image (GKI) with custom device tree
- Integrated out-of-tree drivers & firmware for Wi-Fi/BT and full GPU acceleration via DRM, Mesa 3D.

Certifications

- AWS Certified Solutions Architect – Associate (SAA-C03)
- AWS Certified AI Practitioner (AIF-C01)

Education

Bachelor of Computing Science - Thompson Rivers University

Sept 2019 - Dec 2022

GPA: 4.04/4.33

Kamloops, BC